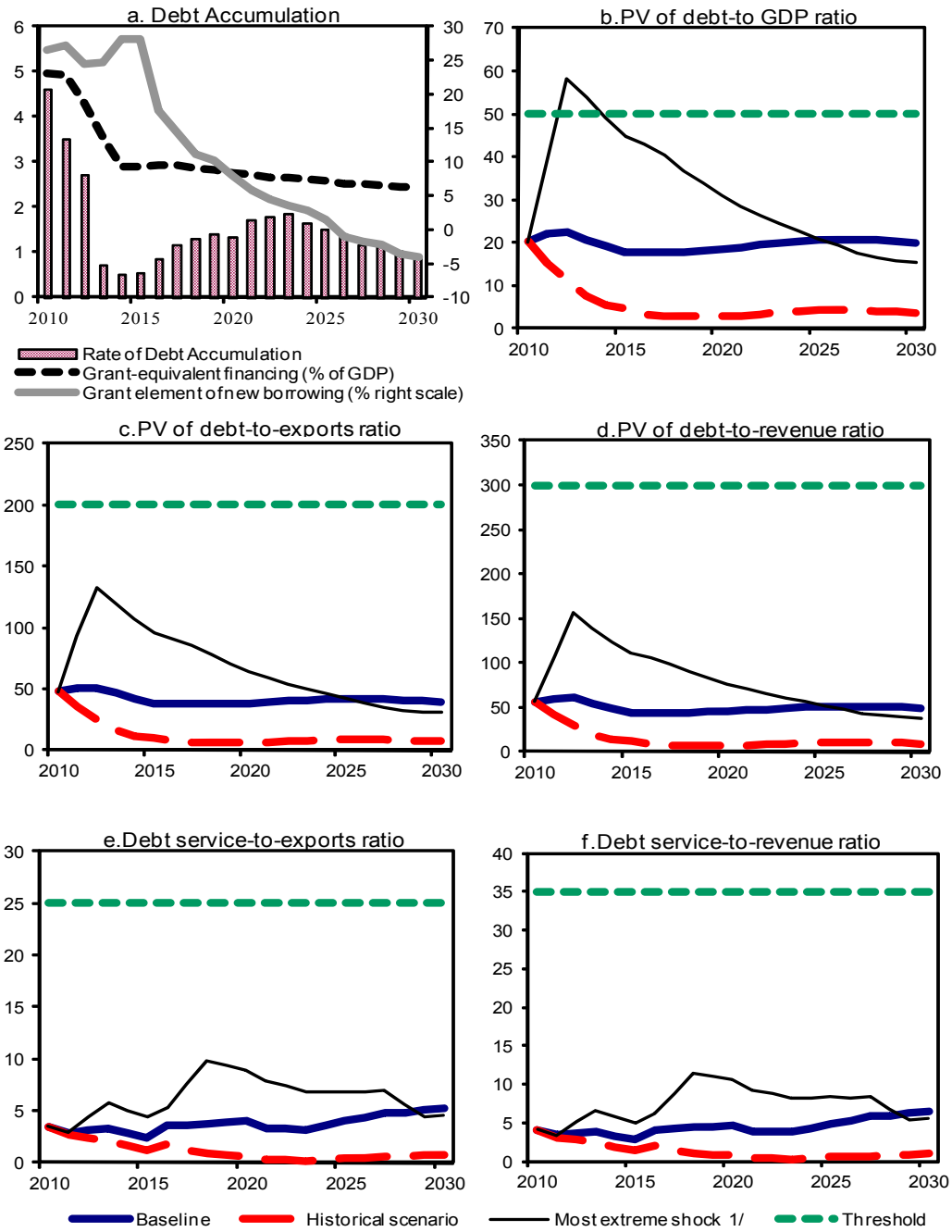
Appendix I. Moldova: Debt Sustainability Analysis Update

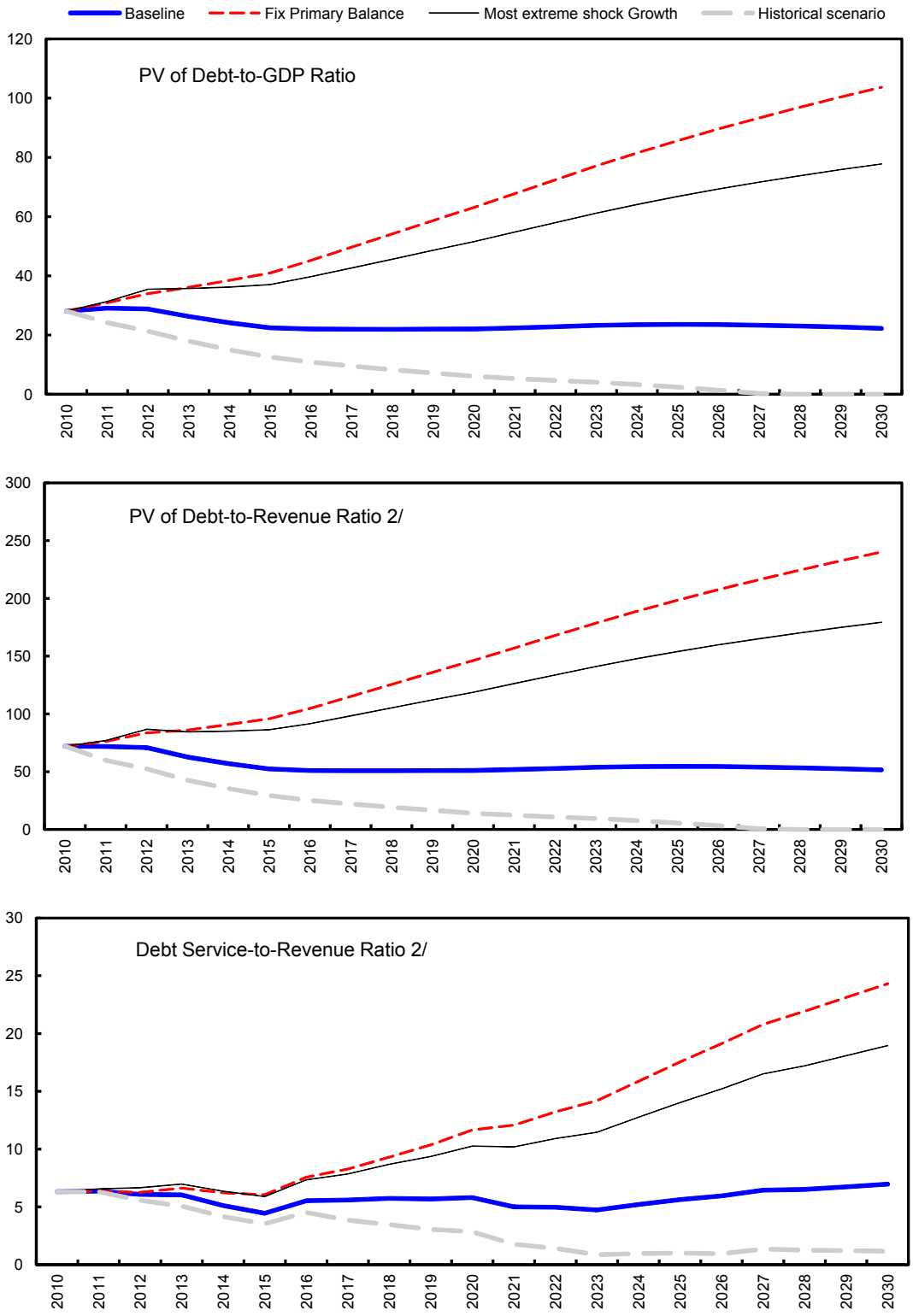
Figure 1A. Moldova: Indicators of Public and Publicly Guaranteed External Debt under Alternative Scenarios, 2010-2030 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2020. In figure b. it corresponds to a Non-debt flows shock; in c. to a Non-debt flows shock; in d. to a Non-debt flows shock; in e. to a Non-debt flows shock and in figure f. to a Non-debt flows shock

Figure 2A.Moldova: Indicators of Public Debt Under Alternative Scenarios, 2010-2030 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2020.
2/ Revenues are defined inclusive of grants.

Table 1A. Moldova: External Debt Sustainability Framework, Baseline Scenario, 2007-2030 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average	Standard Deviation	Projections									
	2007	2008	2009			2010	2011	2012	2013	2014	2015	2010-2015 Average	2020	2030	2016-2030 Average
External debt (nominal) 1/	**62.7**	**55.9**	**68.0**			**74.2**	**77.0**	**77.0**	**73.9**	**70.7**	**67.7**		**61.1**	**52.3**	
o/w public and publicly guaranteed (PPG)	19.9	15.8	22.2			26.2	28.6	29.1	26.6	24.4	22.4		21.8	21.1	
Change in external debt	-1.6	-6.8	12.1			6.2	2.8	0.0	-3.2	-3.2	-3.0		-1.5	-1.1	
Identified net debt-creating flows	-10.8	-12.0	15.2			6.2	4.5	2.6	0.4	-0.3	-1.4		-0.5	-1.0	
Non-interest current account deficit	**14.3**	**15.5**	**7.7**	**5.6**	**6.4**	**8.6**	**9.2**	**8.7**	**7.4**	**6.3**	**5.1**		**5.0**	**5.4**	5.0
Deficit in balance of goods and services	52.5	53.0	36.5			41.1	41.9	41.4	40.4	38.9	37.5		36.4	34.2	
Exports	45.4	41.0	37.1			42.2	43.0	44.1	45.0	45.8	46.6		48.0	50.6	
Imports	97.9	94.0	73.6			83.2	84.9	85.4	85.4	84.7	84.1		84.4	84.8	
Net current transfers (negative = inflow)	-58.8	-55.4	-42.1	-42.5	11.7	-48.4	-48.7	-48.8	-49.4	-49.2	-49.4		-46.7	-41.8	-45.3
o/w official	-1.7	-2.0	-1.8			-2.2	-2.2	-2.2	-2.3	-1.9	-1.7		-1.7	-1.7	
Other current account flows (negative = net inflow)	20.6	18.0	13.3			15.9	16.0	16.1	16.4	16.6	17.0		15.4	13.1	
Net FDI (negative = inflow)	**-12.3**	**-11.7**	**-1.6**	**-6.5**	**3.7**	**-3.9**	**-5.4**	**-5.9**	**-6.5**	**-6.7**	**-6.8**		**-6.6**	**-6.9**	-6.7
Endogenous debt dynamics 2/	**-12.9**	**-15.9**	**9.1**			**1.5**	**0.7**	**-0.2**	**-0.5**	**0.0**	**0.3**		**1.1**	**0.5**	
Contribution from nominal interest rate	1.6	1.3	2.3			3.2	3.1	3.3	3.0	3.4	3.5		3.4	2.5	
Contribution from real GDP growth	-1.5	-3.6	4.1			-1.7	-2.5	-3.5	-3.5	-3.3	-3.2		-2.4	-2.0	
Contribution from price and exchange rate changes	-13.0	-13.6	2.7			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**9.2**	**5.2**	**-3.1**			**0.0**	**-1.6**	**-2.6**	**-3.6**	**-2.9**	**-1.6**		**-1.0**	**-0.1**	
o/w exceptional financing	-0.8	-1.4	-2.5			-3.0	-4.4	-1.9	-2.1	-0.9	-0.9		-0.4	-0.2	
PV of external debt 4/	…	…	63.1			68.2	70.5	70.6	68.1	65.6	63.1		57.6	51.1	
In percent of exports	…	…	170.2			161.8	163.9	160.2	151.4	143.3	135.5		120.0	101.0	
PV of PPG external debt	**…**	**…**	**17.3**			**20.2**	**22.1**	**22.6**	**20.9**	**19.3**	**17.8**		**18.3**	**20.0**	
In percent of exports	**…**	**…**	**46.7**			**47.9**	**51.4**	**51.3**	**46.3**	**42.2**	**38.3**		**38.2**	**39.5**	
In percent of government revenues	**…**	**…**	**46.8**			**56.1**	**59.7**	**60.6**	**53.6**	**48.4**	**44.3**		**45.2**	**49.2**	
Debt service-to-exports ratio (in percent)	**13.7**	**16.6**	**18.6**			**19**	**16**	**17**	**16**	**17**	**18**		**20.4**	**18.9**	
PPG debt service-to-exports ratio (in percent)	**3.9**	**3.3**	**4.2**			**3.5**	**2.9**	**3.2**	**3.3**	**2.8**	**2.4**		**4.0**	**5.3**	
PPG debt service-to-revenue ratio (in percent)	**4.4**	**3.5**	**4.2**			**4.1**	**3.4**	**3.7**	**3.8**	**3.3**	**2.8**		**4.7**	**6.6**	
Total gross financing need (Millions of U.S. dollars)	1261.5	1785.4	2080.4			2103.6	2123.7	2240.4	2203.5	2311.1	2371.3		2849.4	3586.9	
Non-interest current account deficit that stabilizes debt ratio	16.0	22.4	-4.4			2.4	6.4	8.7	10.6	9.5	8.2		6.5	6.5	
Key macroeconomic assumptions															
Real GDP growth (in percent)	3.0	7.8	-6.5	4.7	4.4	2.5	3.6	5.0	5.0	5.0	5.0	4.3	4.0	4.0	4.0
GDP deflator in US dollar terms (change in percent)	25.4	27.6	-4.6	11.8	10.2	-3.1	4.7	4.6	4.9	5.7	5.6	3.7	2.0	2.0	2.0
Effective interest rate (percent) 5/	3.2	2.8	3.7	4.0	0.7	4.7	4.6	4.7	4.3	5.1	5.5	4.8	5.8	5.0	5.8
Growth of exports of G&S (US dollar terms, in percent)	30.9	24.3	-19.3	13.6	14.7	12.9	10.6	12.5	12.5	12.9	12.8	12.4	6.6	6.7	6.7
Growth of imports of G&S (US dollar terms, in percent)	37.5	32.1	-30.1	19.3	19.2	12.4	10.6	10.5	10.2	10.1	10.1	10.6	6.1	6.2	6.1
Grant element of new public sector borrowing (in percent)	…	…	…	…	…	26.5	27.3	24.5	24.6	28.3	28.3	26.6	7.9	-4.1	4.5
Government revenues (excluding grants, in percent of GDP)	39.9	38.9	37.0			36.0	37.0	37.3	38.9	39.9	40.2		40.6	40.6	40.6
Aid flows (in Millions of US dollars) 7/	126.1	127.0	113.8			236.2	293.3	275.4	265.0	246.0	269.4		327.5	536.8	
o/w Grants	79.9	102.8	92.3			162.0	206.3	212.8	220.0	196.0	219.4		297.5	536.8	
o/w Concessional loans	46.2	24.2	21.4			74.3	87.0	62.6	45.0	50.0	50.0		30.0	0.0	
Grant-equivalent financing (in percent of GDP) 8/	…	…	…			5.0	4.9	4.3	3.5	2.9	2.9		2.8	2.4	2.7
Grant-equivalent financing (in percent of external financing) 8/	…	…	…			47.9	57.0	58.5	73.3	74.2	76.0		52.8	47.2	51.7
Memorandum items:															
Nominal GDP (Millions of US dollars)	4401.1	6054.8	5402.9			5367.6	5820.1	6391.6	7043.2	7819.2	8669.3		11645.3	21012.8	
Nominal dollar GDP growth	29.1	37.6	-10.8			-0.7	8.4	9.8	10.2	11.0	10.9	8.3	6.1	6.1	6.1
PV of PPG external debt (in Millions of US dollars)			843.8			1092.6	1280.3	1438.0	1483.3	1519.6	1561.1		2156.7	4241.3	
(PVt-PVt-1)/GDPt-1 (in percent)						4.6	3.5	2.7	0.7	0.5	0.5	2.1	1.3	0.9	1.3

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate
4/ Assumes that PV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 2A. Moldova: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2010-2030
(In percent)

	Projections											
	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	**2020**	2030
PV of debt-to GDP ratio												
Baseline	20	22	23	21	19	18	18	18	18	18	**18**	20
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2010-2030 1/	20	15	11	8	5	5	3	3	3	3	**3**	4
A2. New public sector loans on less favorable terms in 2010-2030 2	20	23	24	23	22	20	20	21	22	22	**23**	28
B. Bound Tests												
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	20	22	22	20	17	15	13	12	12	12	**11**	14
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	20	25	33	30	27	24	23	21	20	19	**17**	14
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	20	22	22	19	17	14	13	12	12	11	**11**	13
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	20	40	58	54	49	45	43	40	37	34	**31**	15
B5. Combination of B1-B4 using one-half standard deviation shocks	20	34	49	45	41	37	35	33	31	28	**26**	15
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	20	29	29	25	22	19	17	16	15	15	**15**	18
PV of debt-to-exports ratio												
Baseline	48	51	51	46	42	38	38	37	38	38	**38**	39
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2010-2030 1/	48	36	25	17	12	10	7	6	6	6	**6**	7
A2. New public sector loans on less favorable terms in 2010-2030 2	48	53	55	51	47	44	44	44	45	47	**48**	54
B. Bound Tests												
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	48	49	46	40	34	29	26	24	23	22	**22**	25
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	48	65	96	85	75	65	61	57	53	49	**46**	34
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	48	49	46	40	34	29	26	24	23	22	**22**	25
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	48	93	132	120	107	96	91	85	78	71	**64**	31
B5. Combination of B1-B4 using one-half standard deviation shocks	48	83	123	111	99	88	83	78	71	65	**60**	32
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	48	49	46	40	34	29	26	24	23	22	**22**	25
PV of debt-to-revenue ratio												
Baseline	56	60	61	54	48	44	43	44	44	45	**45**	49
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2010-2030 1/	56	42	30	19	13	12	8	7	7	7	**7**	9
A2. New public sector loans on less favorable terms in 2010-2030 2	56	62	65	59	54	51	50	52	53	55	**56**	68
B. Bound Tests												
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	56	58	59	50	42	36	33	31	29	28	**28**	34
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	56	68	89	78	67	60	55	52	49	46	**43**	34
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	56	58	58	49	42	35	32	30	29	28	**27**	33
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	56	108	156	138	123	111	105	99	91	83	**76**	38
B5. Combination of B1-B4 using one-half standard deviation shocks	56	92	131	116	102	92	87	82	75	69	**64**	36
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	56	79	76	65	55	47	42	40	38	37	**36**	44

Table 2A. Moldova: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2010-2030 (continued)
(In percent)

	Projections											
	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2030
Debt service-to-exports ratio												
Baseline	3	3	3	3	3	2	4	4	4	4	**4**	5
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2010-2030 1/	3	3	2	2	2	1	2	1	1	1	**1**	1
A2. New public sector loans on less favorable terms in 2010-2030 2	3	3	3	3	3	2	3	2	3	3	**3**	7
B. Bound Tests												
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	3	3	3	3	3	2	3	3	3	3	**3**	4
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	3	3	4	5	4	4	5	6	7	7	**6**	5
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	3	3	3	3	3	2	3	3	3	3	**3**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	3	3	4	6	5	4	5	8	10	9	**9**	5
B5. Combination of B1-B4 using one-half standard deviation shocks	3	3	4	6	5	4	5	7	9	9	**8**	5
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	3	3	3	3	3	2	3	3	3	3	**3**	4
Debt service-to-revenue ratio												
Baseline	4	3	4	4	3	3	4	4	4	5	**5**	7
A. Alternative Scenarios												
A1. Key variables at their historical averages in 2010-2030 1/	4	3	3	2	2	1	2	1	1	1	**1**	1
A2. New public sector loans on less favorable terms in 2010-2030 2	4	3	4	4	4	3	3	3	3	3	**4**	8
B. Bound Tests												
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	4	4	4	4	3	3	4	4	4	4	**4**	5
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	4	3	4	5	4	3	5	5	6	6	**6**	5
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	4	3	4	4	3	3	4	4	4	4	**4**	5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	4	3	5	7	6	5	6	9	12	11	**11**	6
B5. Combination of B1-B4 using one-half standard deviation shocks	4	3	5	6	5	4	6	7	10	9	**9**	5
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	4	5	5	5	4	4	5	5	5	5	**5**	6
Memorandum item:												
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	7	7	7	7	7	7	7	7	7	7	**7**	7

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3A. Moldova: Public Sector Debt Sustainability Framework, Baseline Scenario, 2007-2030
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2007	2008	2009	Average	Standard Deviation	2010	2011	2012	2013	2014	2015	2010-15 Average	2020	2030	2016-30 Average
Public sector debt 1/	26.9	21.3	30.7			34.1	35.6	35.3	32.1	29.3	27.0		25.5	23.4	
o/w foreign-currency denominated	19.9	15.8	22.2			26.2	28.6	29.1	26.6	24.4	22.4		21.8	21.1	
Change in public sector debt	-6.9	-5.6	9.3			3.4	1.6	-0.4	-3.2	-2.8	-2.2		-0.2	-0.7	
Identified debt-creating flows	-8.2	-6.4	10.9			2.1	1.3	-1.4	-3.2	-2.7	-2.4		-1.2	-1.2	
Primary deficit	-1.0	-0.2	5.4	-2.1	3.2	4.3	2.6	0.8	-0.6	-0.8	-0.7	0.9	-0.7	-0.8	-0.8
Revenue and grants	41.7	40.6	38.7			39.0	40.5	40.7	42.0	42.4	42.8		43.2	43.2	
of which: grants	1.8	1.7	1.7			3.0	3.5	3.3	3.1	2.5	2.5		2.6	2.6	
Primary (noninterest) expenditure	40.8	40.4	44.1			43.3	43.1	41.5	41.4	41.6	42.1		42.4	42.4	
Automatic debt dynamics	-6.9	-4.2	5.5			-1.7	-1.1	-2.0	-2.5	-2.1	-2.0		-0.6	-0.4	
Contribution from interest rate/growth differential	-5.9	-5.3	3.6			0.6	-1.4	-2.1	-2.1	-2.1	-1.9		-0.6	-0.4	
of which: contribution from average real interest rate	-4.9	-3.4	2.1			1.4	-0.2	-0.4	-0.4	-0.6	-0.5		0.4	0.5	
of which: contribution from real GDP growth	-1.0	-1.9	1.5			-0.7	-1.2	-1.7	-1.7	-1.5	-1.4		-1.0	-0.9	
Contribution from real exchange rate depreciation	-1.0	1.1	1.9			-2.4	0.3	0.0	-0.4	0.1	0.0		…	…	
Other identified debt-creating flows	-0.3	-1.9	0.0			-0.5	-0.2	-0.2	-0.2	0.2	0.2		0.2	0.1	
Privatization receipts (negative)	-0.3	-1.9	-0.4			-0.9	-0.6	-0.5	-0.5	-0.1	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.4			0.4	0.3	0.3	0.3	0.3	0.2		0.2	0.1	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	1.4	0.8	-1.6			1.3	0.3	1.0	0.0	-0.1	0.2		1.0	0.5	
Other Sustainability Indicators															
PV of public sector debt	7.0	5.6	25.8			28.1	29.1	28.8	26.3	24.1	22.5		22.0	22.3	
o/w foreign-currency denominated	0.0	0.0	17.3			20.2	22.1	22.6	20.9	19.3	17.8		18.3	20.0	
o/w external	…	…	17.3			20.2	22.1	22.6	20.9	19.3	17.8		18.3	20.0	
PV of contingent liabilities (not included in public sector debt)	…	…	…			…	…	…	…	…	…		…	…	
Gross financing need 2/	4.8	4.9	10.6			13.6	11.7	9.3	7.4	6.3	5.5		5.2	4.3	
PV of public sector debt-to-revenue and grants ratio (in percent)	16.8	13.8	66.7			72.1	71.8	70.8	62.7	57.0	52.5		51.0	51.5	
PV of public sector debt-to-revenue ratio (in percent)	17.6	14.4	69.7			78.1	78.7	77.2	67.7	60.6	55.8		54.2	54.8	
o/w external 3/	…	…	46.8			56.1	59.7	60.6	53.6	48.4	44.3		45.2	49.2	
Debt service-to-revenue and grants ratio (in percent) 4/	6.9	6.0	6.8			6.3	6.4	6.1	6.0	5.1	4.5		5.8	7.0	
Debt service-to-revenue ratio (in percent) 4/	7.2	6.3	7.1			6.8	7.0	6.6	6.5	5.4	4.7		6.2	7.4	
Primary deficit that stabilizes the debt-to-GDP ratio	5.9	5.4	-3.9			0.9	1.0	1.2	2.6	2.0	1.5		-0.5	-0.2	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	3.0	7.8	-6.5	4.7	4.4	2.5	3.6	5.0	5.0	5.0	5.0	4.3	4.0	4.0	4.0
Average nominal interest rate on forex debt (in percent)	1.9	1.8	2.0	3.2	1.1	1.7	1.2	1.3	1.3	1.3	1.4	1.4	2.2	3.6	2.6
Average real interest rate on domestic debt (in percent)	-3.8	5.4	15.8	1.3	6.8	3.8	8.5	7.6	10.0	8.8	8.7	7.9	8.7	8.7	8.7
Real exchange rate depreciation (in percent, + indicates depreciation)	-5.1	7.3	10.8	0.3	6.6	-10.5	…	…	…	…	…	…	…	…	…
Inflation rate (GDP deflator, in percent)	15.9	9.2	2.0	12.2	6.6	9.0	5.5	5.0	4.5	4.0	4.0	5.3	4.0	4.0	4.0
Growth of real primary spending (deflated by GDP deflator, in percent)	0.1	0.1	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.1	0.1	0.0	0.0	0.0	0.0
Grant element of new external borrowing (in percent)	…	…	…	…	…	26.5	27.3	24.5	24.6	28.3	28.3	26.6	7.9	-4.1	…

Sources: Country authorities; and staff estimates and projections.
1/ General government gross debt, excluding debt of state-owned enterprises.
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
3/ Revenues excluding grants.
4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4A. Moldova: Sensitivity Analysis for Key Indicators of Public Debt, 2010-2030

	Projections							
	2010	2011	2012	2013	2014	2015	2020	2030
PV of Debt-to-GDP Ratio								
Baseline	28	29	29	26	24	22	22	22
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	28	24	21	18	15	13	6	0
A2. Primary balance is unchanged from 2010	28	31	34	36	38	41	63	104
A3. Permanently lower GDP growth 1/	28	30	30	29	29	29	44	106
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2011-2012	28	31	35	36	36	37	52	78
B2. Primary balance is at historical average minus one standard deviations in 2011-2012	28	28	28	25	23	22	21	22
B3. Combination of B1-B2 using one half standard deviation shocks	28	26	26	25	24	24	30	43
B4. One-time 30 percent real depreciation in 2011	28	37	36	33	30	28	28	32
B5. 10 percent of GDP increase in other debt-creating flows in 2011	28	39	38	35	32	30	29	28
PV of Debt-to-Revenue Ratio 2/								
Baseline	72	72	71	63	57	53	51	52
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	72	60	52	43	35	29	14	0
A2. Primary balance is unchanged from 2010	72	76	84	86	91	96	146	240
A3. Permanently lower GDP growth 1/	72	73	75	69	67	67	102	242
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2011-2012	72	77	87	84	85	86	119	179
B2. Primary balance is at historical average minus one standard deviations in 2011-2012	72	68	68	60	55	51	49	50
B3. Combination of B1-B2 using one half standard deviation shocks	72	65	63	59	57	56	70	99
B4. One-time 30 percent real depreciation in 2011	72	92	89	79	72	66	65	74
B5. 10 percent of GDP increase in other debt-creating flows in 2011	72	96	94	84	77	71	67	65
Debt Service-to-Revenue Ratio 2/								
Baseline	6	6	6	6	5	4	6	7
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	6	6	6	5	4	4	3	1
A2. Primary balance is unchanged from 2010	6	6	6	7	6	6	12	24
A3. Permanently lower GDP growth 1/	6	6	6	6	6	5	9	22
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2011-2012	6	7	7	7	6	6	10	19
B2. Primary balance is at historical average minus one standard deviations in 2011-2012	6	6	6	6	5	4	6	7
B3. Combination of B1-B2 using one half standard deviation shocks	6	6	6	6	5	5	6	11
B4. One-time 30 percent real depreciation in 2011	6	7	8	8	7	6	8	12
B5. 10 percent of GDP increase in other debt-creating flows in 2011	6	6	7	7	6	5	8	9

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.